

23 Chemical energetics

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

At A Level, entropy measures dispersal of energy and arrangements; $\Delta G = \Delta H - T\Delta S$ predicts feasible direction at specified temperature under standard conditions. A negative ΔG does not ensure fast reaction. Standard enthalpy changes may be derived from formation data using products minus reactants.

Worked example: follow the reasoning

For $\Delta H = +40 \text{ kJ mol}^{-1}$ and $\Delta S = +0.20 \text{ kJ mol}^{-1} \text{ K}^{-1}$, ΔG at 300 K = $40 - 60 = -20 \text{ kJ mol}^{-1}$.

Quick check · multiple choice

A What criterion of ΔG supports feasibility at specified conditions?

[1]

A $\Delta G > 0$ always

B $\Delta G = \Delta H$ only

C $\Delta G < 0$

D ΔG depends only on catalyst

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A process has $\Delta H = +75 \text{ kJ mol}^{-1}$ and $\Delta S = +0.150 \text{ kJ mol}^{-1} \text{ K}^{-1}$. Find threshold temperature for $\Delta G < 0$.

[3]

Final answer: _____

Answers and explanation

Quick check: C. $\Delta G < 0$

Question 1 [3 marks]

Set $\Delta G = 0$, $T = \Delta H/\Delta S$ [1] = $75/0.150 = 500$ K [1]; feasible by ΔG criterion above 500 K [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.